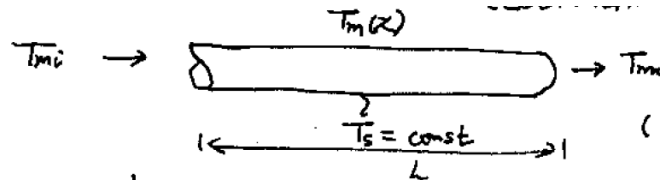


18. Log Mean Temperature Difference (LMTD)

Internal flow HT with const. temp. BC



[Mean temperature variation]

$T_s = \text{const}$ & Fully developed flow ($h = \bar{h} = \text{const}$)

$$\frac{dT_m}{dx} = \frac{Ph}{\dot{m}c_p}(T_s - T_m) \rightarrow \frac{T_s - T_m(x)}{T_s - T_{mi}} = \exp\left(-\frac{Phx}{\dot{m}c_p}\right)$$

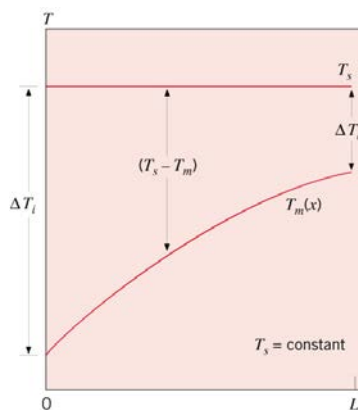
$$\text{If } h \neq \text{const} \rightarrow \frac{T_s - T_m(x)}{T_s - T_{mi}} = \exp\left(-\frac{P\bar{h}x}{\dot{m}c_p}\right)$$

[Heat transfer]

- Total HT: $q_{\text{conv}} = \dot{m}C_p(T_{mo} - T_{mi})$
- Local HT: $q'' = h[T_s - T_m(x)]$
- How about $q_{\text{conv}} = \bar{h}A\bar{\Delta T}$?

Note that ΔT is a function of x . Then, what is the average ΔT , i.e., $\bar{\Delta T}$?

How to average the temperature difference?



$$q_{\text{conv}} = \bar{h}A\bar{\Delta T} = \dot{m}C_p(T_{mo} - T_{mi})$$

$$\frac{T_s - T_{mo}}{T_s - T_{mi}} = \exp\left(-\frac{P\bar{h}L}{\dot{m}c_p}\right) \rightarrow \ln \frac{T_s - T_{mo}}{T_s - T_{mi}} = -\frac{\bar{h}A}{\dot{m}c_p} \rightarrow \bar{h}A = \dot{m}c_p \ln \frac{T_s - T_{mi}}{T_s - T_{mo}}$$

Substitution of the above $\bar{h}A$ into $\bar{h}A\bar{\Delta T} = \dot{m}C_p(T_{mo} - T_{mi})$ yields

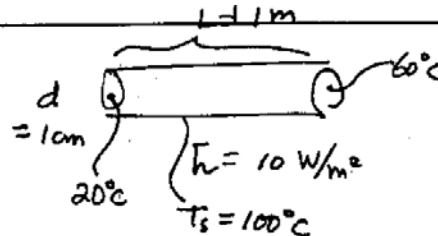
$$\bar{\Delta T} = \frac{T_{mo} - T_{mi}}{\ln \frac{T_s - T_{mi}}{T_s - T_{mo}}} = \frac{(T_s - T_{mi}) - (T_s - T_{mo})}{\ln \frac{T_s - T_{mi}}{T_s - T_{mo}}} = \frac{\Delta T_i - \Delta T_o}{\ln \frac{\Delta T_i}{\Delta T_o}} = \Delta T_{lm}$$

($\Delta T \equiv T_s - T_m$)

$$\Delta T_{lm} = \frac{\Delta T_i - \Delta T_o}{\ln \frac{\Delta T_i}{\Delta T_o}}$$

 LMTD (log-mean temperature difference)

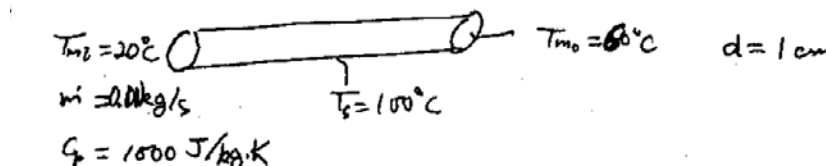
Ex1)



$Q = ?$
 $Q = \dot{m}C_p(T_{mo} - T_{mi})$
 $Q = \bar{h}A\Delta T_{lm}$
 $= \bar{h}A \frac{\Delta T_{mo} - \Delta T_{mi}}{\ln \left(\frac{\Delta T_{mo}}{\Delta T_{mi}} \right)} = \bar{h} \pi d L \frac{40 - 20}{\ln \left(\frac{40}{20} \right)}$
 $\Delta T_{lm} \approx 57.7^\circ\text{C}$

$\frac{T_s - T_{mi}}{T_s - T_{mo}} = \exp\left(-\frac{\bar{h}A}{\dot{m}C_p}\right)$

Ex2)



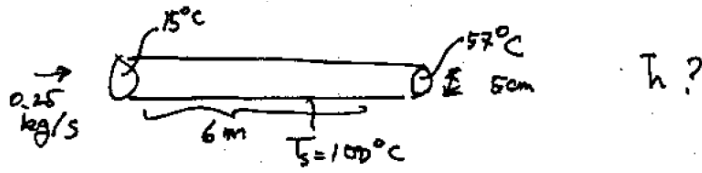
$T_{mi} = 20^\circ\text{C}$, $T_{mo} = 60^\circ\text{C}$, $d = 1\text{ cm}$
 $\dot{m} = 0.01\text{ kg/s}$, $T_s = 100^\circ\text{C}$
 $C_p = 1000\text{ J/kg.K}$
 $\bar{h} = 10\text{ W/m}^2$, $L = ?$

$Q = \dot{m}C_p(T_{mo} - T_{mi}) = \bar{h}(\pi d L)\Delta T_{lm}$
 $0.01 \times 1000 \times (60 - 20) = 10 \times (\pi \times 0.01 \times L) \times \frac{40 - 20}{\ln \left(\frac{40}{20} \right)}$
 $\therefore L = 33\text{ m}$

Alt. Sol

$$\frac{T_s - T_m(\text{in})}{T_s - T_{mi}} = \exp\left(-\frac{PL\bar{h}}{\dot{m}c_p}\right)$$

Ex3)



$$q = \bar{h} A \Delta T_{\text{em}} = \dot{m} c_p (T_{\text{mo}} - T_{\text{mi}})$$

$$\bar{h} = \frac{\dot{m} c_p}{\pi d L} \frac{T_{\text{mo}} - T_{\text{mi}}}{\Delta T_{\text{em}}} \quad \Delta T_{\text{em}} = \frac{43 - 85}{\ln(43/85)} = 61.6^\circ\text{C}$$